📘 Modul Materi:
Desain Website Grocery Store Pontianak

HTML5 + CSS3 + JavaScript (Vanilla) — Responsive — Dari Nol Sampai Publish

📋 Daftar Isi Modul

1. Orientasi Proyek & Persiapan
2. Membuat Struktur Folder & File
3. Kerangka HTML5 (Skeleton Semua Halaman)
4. CSS Dasar: Variabel, Reset, Layout Global
5. Responsive Design (Mobile-First)
6. Halaman Beranda (index.html)
7. Halaman Katalog Produk + Data Produk
8. Halaman Detail Produk
9. Keranjang Belanja (localStorage)
10. Halaman Checkout + Validasi Form
11. Halaman Kontak & Tentang Kami
12. Testing & Debugging
13. SEO Dasar & Publish Online

BAB 1 — Orientasi Proyek & Persiapan

1.1 Tujuan Pembelajaran

Di akhir modul, kamu bisa:

- Membuat struktur proyek web yang rapi dan profesional.
- Menyusun halaman web grocery store sesuai standar (product listing, cart, checkout, kontak, dll).
- Menerapkan responsive design untuk desktop, tablet, dan mobile.
- Menggunakan JavaScript untuk interaktivitas (keranjang, filter, form).
- Memublish website agar bisa diakses publik.

1.2 Standar Halaman Web Grocery Store

Setiap toko kelontong/online grocery wajib punya halaman ini:

- **Beranda (Home)** — hero, kategori, produk unggulan, promo, keunggulan.
- **Katalog Produk (Products)** — daftar produk + pencarian + filter kategori.
- **Detail Produk (Product Detail)** — gambar, harga, deskripsi, tombol add-to-cart.
- **Keranjang (Cart)** — item, ubah jumlah, hapus, total harga.
- **Checkout** — data pembeli, alamat pengiriman, metode bayar, ringkasan pesanan.
- **Tentang Kami (About)** — profil toko.
- **Kontak (Contact)** — alamat Pontianak, WhatsApp, jam buka, form pesan.

1.3 Alat yang Dipakai

- VS Code (teks editor)
- Browser Chrome/Firefox (testing + DevTools)
- Tidak perlu install apa pun lagi — semua kode berjalan langsung di browser.

BAB 2 — Membuat Struktur Folder & File

Langkah 1 — Buat folder utama bernama `grocery-pontianak` di lokasi mana pun (misal `D:\latihan`).

Langkah 2 — Buat subfolder dan file di dalamnya sesuai struktur ini:

```
grocery-pontianak/
│
├── index.html              ← Beranda
├── products.html           ← Katalog produk
├── product-detail.html     ← Detail produk
├── cart.html               ← Keranjang
├── checkout.html           ← Checkout
├── about.html              ← Tentang kami
├── contact.html            ← Kontak
│
├── css/
│   ├── reset.css           ← Reset style browser
│   ├── style.css           ← Style utama
│   └── responsive.css      ← Media query responsive
│
├── js/
│   ├── data.js             ← Data produk (array)
│   ├── app.js              ← Fitur global (nav mobile, badge cart)
│   ├── products.js         ← Render produk + search/filter
│   ├── product-detail.js   ← Tampilkan produk dari URL ?id=
│   ├── cart.js             ← Logika keranjang
│   └── validation.js       ← Validasi form checkout/kontak
│
├── assets/
│   ├── images/             ← Foto produk, logo, banner
│   └── icons/              ← Ikon (opsional, pakai emoji/svg)
│
└── README.md               ← Deskripsi proyek
```

Langkah 3 — Buka folder `grocery-pontianak` di VS Code (File → Open Folder).

Langkah 4 — Buat semua file di atas (klik kanan → New File). Kosong dulu, kita isi bertahap.

> 💡 **Tips:** Selalu gunakan path relatif (`css/style.css`, bukan `/css/style.css`) supaya website tetap jalan saat di-upload ke hosting.

BAB 3 — Kerangka HTML5 Semua Halaman

3.1 Template Dasar

Semua halaman memakai kerangka yang sama. Contoh untuk `index.html`:

```
<!DOCTYPE html>
<html lang="id">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Toko grocery online Pontianak - belanja kebutuhan sembako, sayur, buah segar langsung
dari rumah.">
  <title>Beranda - Grosir Segar Pontianak</title>
  <link rel="stylesheet" href="css/reset.css">
  <link rel="stylesheet" href="css/style.css">
  <link rel="stylesheet" href="css/responsive.css">
</head>
<body>

  <!-- HEADER -->
  <header class="header">
    <div class="container header__inner">
      <a href="index.html" class="logo">🛒 Grosir<span>Segar</span> Pontianak</a>

      <button class="nav-toggle" id="navToggle" aria-label="Buka menu">☰</button>

      <nav class="nav" id="navMenu">
        <a href="index.html" class="nav__link active">Beranda</a>
        <a href="products.html" class="nav__link">Produk</a>
        <a href="about.html" class="nav__link">Tentang</a>
        <a href="contact.html" class="nav__link">Kontak</a>
        <a href="cart.html" class="nav__link cart-link">🛒 Keranjang <span class="cart-badge" id="cartBadge">0</span></a>
      </nav>
    </div>
  </header>

  <!-- KONTEN UTAMA (diisi per halaman) -->
  <main>
    <section class="container">
      <h1>Isi halaman di sini</h1>
    </section>
  </main>

  <!-- FOOTER -->
  <footer class="footer">
    <div class="container footer__grid">
      <div>
        <h3>🛒 Grosir Segar Pontianak</h3>
        <p>Kebutuhan dapur lengkap, harga bersahabat, pengiriman cepat di seluruh Pontianak.</p>
      </div>
      <div>
        <h3>Menu</h3>
        <a href="products.html">Produk</a>
        <a href="about.html">Tentang Kami</a>
        <a href="contact.html">Kontak</a>
      </div>
      <div>
        <h3>Kontak</h3>
        <p>Jl. Contoh Raya No. 12, Pontianak</p>
        <p>WhatsApp: 0812-xxxx-xxxx</p>
        <p>Setiap hari: 07.00 – 21.00 WIB</p>
      </div>
    </div>
    <div class="footer__bottom">
      <p>© 2025 Grosir Segar Pontianak. All rights reserved.</p>
    </div>
  </footer>

  <script src="js/data.js"></script>
  <script src="js/app.js"></script>
</body>
</html>
```

Langkah praktik:

1. Simpan template di atas sebagai `index.html`.

2. Copy-paste ke semua file halaman (`products.html` , `cart.html` , dll).
3. Ubah hanya: `<title>` , meta description, bagian `<main>` , dan class `active` pada nav.
4. Tambahkan `<script>` khusus halaman sebelum `</body>` .

> ⚠️ **Aturan urutan script:** `data.js` (data) → file fitur halaman → `app.js` (global). Jangan dibalik!

BAB 4 — CSS Dasar: Reset, Variabel, Layout Global

4.1 `css/reset.css`

```
* { margin: 0; padding: 0; box-sizing: border-box; }
html { scroll-behavior: smooth; }
body { font-family: 'Segoe UI', Arial, sans-serif; line-height: 1.6; color: #222; }
img { max-width: 100%; display: block; }
a { text-decoration: none; color: inherit; }
ul { list-style: none; }
button { cursor: pointer; font-family: inherit; }
input, textarea, select { font-family: inherit; }
```

4.2 `css/style.css` — Variabel & Layout Dasar

```
/* ===== CSS VARIABLES ===== */
:root {
  --primary: #2e8b57;        /* hijau segar */
  --primary-dark: #1f6b41;
  --accent: #ff9800;         /* oranye promo */
  --bg: #f7faf7;
  --white: #ffffff;
  --text: #222222;
  --muted: #6b7280;
  --border: #e5e7eb;
  --radius: 10px;
  --shadow: 0 2px 8px rgba(0,0,0,.08);
  --max-width: 1100px;
}

.container {
  width: 100%;
  max-width: var(--max-width);
  margin: 0 auto;
  padding: 0 1rem;
}

main { min-height: 60vh; padding: 1.5rem 0; }
section { margin-bottom: 2.5rem; }

h1, h2, h3 { margin-bottom: .75rem; color: var(--text); }
h2 { font-size: 1.6rem; border-left: 5px solid var(--primary); padding-left: .5rem; }
p  { margin-bottom: .75rem; }
```

4.3 Header, Navigasi, Footer

```
/* ===== HEADER ===== */
.header {
  background: var(--white);
  box-shadow: var(--shadow);
  position: sticky; top: 0; z-index: 100;
}
.header__inner {
  display: flex; align-items: center; justify-content: space-between;
  padding-top: .8rem; padding-bottom: .8rem;
}
.logo { font-size: 1.25rem; font-weight: 700; }
.logo span { color: var(--primary); }

.nav { display: flex; gap: 1.25rem; align-items: center; }
.nav__link { font-weight: 500; padding: .35rem 0; position: relative; }
.nav__link:hover, .nav__link.active { color: var(--primary); }

.cart-link { background: var(--primary); color: #fff !important;
  padding: .4rem .8rem !important; border-radius: var(--radius); }
.cart-badge {
  background: var(--accent); color: #fff; border-radius: 50%;
  padding: 0 .45rem; font-size: .8rem; margin-left: .25rem;
}

/* ===== FOOTER ===== */
.footer { background: #123524; color: #d9e5dd; margin-top: 3rem; }
.footer__grid {
  display: grid; grid-template-columns: repeat(3, 1fr);
  gap: 1.5rem; padding-top: 2rem; padding-bottom: 2rem;
}
.footer h3 { color: #fff; font-size: 1.05rem; margin-bottom: .5rem; }
.footer a { display: block; padding: .15rem 0; }
.footer a:hover { color: var(--accent); }
.footer__bottom {
  border-top: 1px solid #2c5440; text-align: center;
  padding: 1rem; font-size: .9rem;
}
```

4.4 Komponen: Tombol, Kartu Produk, Form

```
/* ===== TOMBOL ===== */
.btn {
  display: inline-block; background: var(--primary); color: #fff;
  padding: .65rem 1.4rem; border: none; border-radius: var(--radius);
  font-size: 1rem; font-weight: 600; transition: .2s;
}
.btn:hover { background: var(--primary-dark); }
.btn--outline { background: transparent; border: 2px solid var(--primary); color: var(--primary); }
.btn--accent { background: var(--accent); }
.btn--block { display: block; width: 100%; text-align: center; }

/* ===== KARTU PRODUK ===== */
.product-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(220px, 1fr));
  gap: 1.25rem;
}
.card {
  background: var(--white); border: 1px solid var(--border);
  border-radius: var(--radius); overflow: hidden; box-shadow: var(--shadow);
  display: flex; flex-direction: column; transition: transform .2s;
}
.card:hover { transform: translateY(-4px); }
.card__img { height: 150px; object-fit: cover; width: 100%; background: #eef5ef; }
.card__body { padding: .9rem; flex: 1; display: flex; flex-direction: column; }
.card__cat { font-size: .78rem; color: var(--muted); text-transform: uppercase; }
.card__title { font-size: 1rem; margin: .25rem 0; }
.card__price { color: var(--primary); font-weight: 700; font-size: 1.1rem; margin-bottom: .75rem; }
.card__body .btn { margin-top: auto; }

.badge-promo {
  background: var(--accent); color: #fff; font-size: .75rem;
  padding: .15rem .5rem; border-radius: 999px;
}

/* ===== FORM ===== */
.form-group { margin-bottom: 1rem; }
.form-group label { display: block; font-weight: 600; margin-bottom: .35rem; }
.form-group input, .form-group textarea, .form-group select {
  width: 100%; padding: .65rem .8rem; border: 1px solid var(--border);
  border-radius: var(--radius); font-size: 1rem;
}
.form-group input:focus, .form-group textarea:focus {
  outline: 2px solid var(--primary); border-color: var(--primary);
}
.error-msg { color: #dc2626; font-size: .85rem; margin-top: .25rem; display: none; }
.form-group.invalid input, .form-group.invalid textarea { border-color: #dc2626; }
.form-group.invalid .error-msg { display: block; }

/* ===== ALERT ===== */
.alert-success {
  background: #dcfce7; border: 1px solid #2e8b57; color: #166534;
  padding: 1rem; border-radius: var(--radius); margin-bottom: 1rem;
}
```

BAB 5 — Responsive Design (Mobile-First)

5.1 Prinsip

- Desain dulu untuk mobile (1 kolom), lalu tambah lebar layar dengan `@media (min-width: ...)`.
- Breakpoint yang dipakai:
  - ≤767px → mobile (default)

- ≥768px → tablet
- ≥1024px → desktop

5.2 `css/responsive.css`

```
/* ====== MOBILE (default, <= 767px) ====== */
.nav-toggle {
  display: block; background: none; border: none;
  font-size: 1.6rem; color: var(--text);
}

/* Menu nav disembunyikan dulu di mobile, muncul saat diklik */
.nav {
  position: absolute; top: 100%; left: 0; right: 0;
  background: var(--white); flex-direction: column;
  align-items: flex-start; padding: 1rem;
  gap: .75rem; box-shadow: var(--shadow);
  display: none;
}
.nav.show { display: flex; }

.footer__grid { grid-template-columns: 1fr; }

/* Toolbar produk (search/filter) menumpuk */
.shop-toolbar { flex-direction: column; gap: .75rem; }
.shop-toolbar input, .shop-toolbar select { width: 100%; }

/* Tabel keranjang jadi kartu di mobile */
.table-wrap { overflow-x: auto; }

/* ====== TABLET (>= 768px) ====== */
@media (min-width: 768px) {
  .nav-toggle { display: none; }
  .nav { display: flex; flex-direction: row; position: static;
         padding: 0; box-shadow: none; }
  .footer__grid { grid-template-columns: repeat(2, 1fr); }
  h1 { font-size: 2rem; }
}

/* ====== DESKTOP (>= 1024px) ====== */
@media (min-width: 1024px) {
  .footer__grid { grid-template-columns: repeat(3, 1fr); }
  main { padding: 2.5rem 0; }
}
```

5.3 Elemen Wajib Responsive

- `<meta name="viewport" content="width=device-width, initial-scale=1.0">` — wajib ada di setiap halaman.
- `img { max-width: 100% }` — gambar tidak meluber.
- Container max-width + padding — konten tidak menempel di tepi layar.
- Tombol/area klik minimal ±44px agar nyaman disentuh.

BAB 6 — Halaman Beranda (index.html)

Isi `<main>` dengan bagian-bagian berikut:

```
<!-- HERO -->
<section class="hero">
  <div class="container">
    <h1>Belanja Kebutuhan Dapur Tanpa Ribet 🌿</h1>
    <p>Sembako, sayur, buah segar, dan kebutuhan harian lainnya diantar ke rumah Anda di seluruh Pontianak.</p>
    <div class="hero__actions">
      <a href="products.html" class="btn">Belanja Sekarang</a>
      <a href="about.html" class="btn btn--outline">Tentang Kami</a>
    </div>
  </div>
</section>

<!-- KATEGORI -->
<section class="container">
  <h2>Belanja per Kategori</h2>
  <div class="category-grid">
    <a href="products.html?cat=sembako" class="category-card">🌾 Sembako</a>
    <a href="products.html?cat=sayur"    class="category-card">🥬 Sayur</a>
    <a href="products.html?cat=buah"     class="category-card">🍎 Buah</a>
    <a href="products.html?cat=minuman"  class="category-card">🧃 Minuman</a>
  </div>
</section>

<!-- PRODUK UNGGULAN (diisi JS) -->
<section class="container">
  <h2>Produk Unggulan</h2>
  <div class="product-grid" id="featuredProducts"></div>
</section>

<!-- KEUNGGULAN -->
<section class="container">
  <h2>Kenapa Belanja di Sini?</h2>
  <div class="feature-grid">
    <div class="card"><div class="card__body"><h3>🚚 Pengiriman Cepat</h3><p>Same-day delivery area Pontianak Kota.</p>
</div></div>
    <div class="card"><div class="card__body"><h3>✅ Produk Segar</h3><p>Garansi segar atau uang kembali.</p></div></div>
    <div class="card"><div class="card__body"><h3>💰 Harga Terjangkau</h3><p>Harga grosir untuk semua pelanggan.</p>
</div></div>
  </div>
</section>
```

Tambahkan CSS pendukung di `style.css`:

```
/* HERO */
.hero {
  background: linear-gradient(135deg, var(--primary), var(--primary-dark));
  color: #fff; text-align: center; padding: 3.5rem 1rem;
  border-radius: var(--radius); margin-bottom: 2.5rem;
}
.hero h1 { color: #fff; font-size: 1.8rem; margin-bottom: .75rem; }
.hero p { max-width: 600px; margin: 0 auto 1.25rem; }
.hero__actions { display: flex; gap: .75rem; justify-content: center; flex-wrap: wrap; }
.hero .btn--outline { border-color: #fff; color: #fff; }

/* KATEGORI */
.category-grid {
  display: grid; grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
  gap: 1rem;
}
.category-card {
  background: var(--white); border: 1px solid var(--border);
  border-radius: var(--radius); text-align: center;
  padding: 1.25rem .75rem; font-weight: 600; transition: .2s;
}
.category-card:hover { border-color: var(--primary); color: var(--primary); }

/* FEATURE */
.feature-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(220px, 1fr)); gap: 1rem; }
```

BAB 7 — Katalog Produk (products.html + data.js + products.js)

7.1 Struktur Halaman

```
<main class="container">
  <h1>Semua Produk</h1>

  <!-- Toolbar: search + filter + sort -->
  <div class="shop-toolbar" style="display:flex; margin-bottom:1.5rem;">
    <input type="search" id="searchInput" placeholder="Cari produk..." style="padding:.6rem; border:1px solid var(--
border); border-radius:10px;">
    <select id="categoryFilter" style="padding:.6rem; border:1px solid var(--border); border-radius:10px;">
      <option value="all">Semua Kategori</option>
      <option value="sembako">Sembako</option>
      <option value="sayur">Sayur</option>
      <option value="buah">Buah</option>
      <option value="minuman">Minuman</option>
    </select>
    <select id="sortSelect" style="padding:.6rem; border:1px solid var(--border); border-radius:10px;">
      <option value="default">Urutkan</option>
      <option value="low">Harga Terendah</option>
      <option value="high">Harga Tertinggi</option>
      <option value="name">Nama A-Z</option>
    </select>
  </div>

  <div class="product-grid" id="productList"></div>
  <p id="emptyMessage" style="display:none;">😕 Produk tidak ditemukan.</p>
</main>
```

7.2 `js/data.js` — Data Produk

```
const PRODUCTS = [
  { id: 1, name: "Beras Pandan Wangi 5 kg", category: "sembako", price: 72000, promo: false,
    unit: "5 kg", stock: 25,
    image: "https://placehold.co/400x300?text=Beras+5kg",
    desc: "Beras pulen khas Kalimantan, cocok untuk nasi harian keluarga." },
  { id: 2, name: "Minyak Goreng 2 L", category: "sembako", price: 38000, promo: true,
    unit: "2 liter", stock: 40,
    image: "https://placehold.co/400x300?text=Minyak+Goreng",
    desc: "Minyak goreng kelapa sawit murni, jernih dan hemat." },
  { id: 3, name: "Gula Pasir 1 kg", category: "sembako", price: 17000, promo: false,
    unit: "1 kg", stock: 60,
    image: "https://placehold.co/400x300?text=Gula+Pasir",
    desc: "Gula pasir putih halus bersertifikat SNI." },
  { id: 4, name: "Telur Ayam Negeri", category: "sembako", price: 28000, promo: false,
    unit: "1 kg (±16 butir)", stock: 30,
    image: "https://placehold.co/400x300?text=Telur+Ayam",
    desc: "Telur ayam segar dari peternakan lokal Pontianak." },
  { id: 5, name: "Bayam Segar 1 ikat", category: "sayur", price: 5000, promo: true,
    unit: "1 ikat", stock: 50,
    image: "https://placehold.co/400x300?text=Bayam",
    desc: "Bayam segar dipetik pagi hari, cocok untuk sayur bening." },
  { id: 6, name: "Kangkung Segar", category: "sayur", price: 4000, promo: false,
    unit: "1 ikat", stock: 45,
    image: "https://placehold.co/400x300?text=Kangkung",
    desc: "Kangkung muda segar dan renyah." },
  { id: 7, name: "Cabai Merah Keriting", category: "sayur", price: 35000, promo: false,
    unit: "500 gram", stock: 20,
    image: "https://placehold.co/400x300?text=Cabai+Merah",
    desc: "Cabai merah segar, pedas mantap untuk sambal." },
  { id: 8, name: "Jeruk Pontianak", category: "buah", price: 25000, promo: true,
    unit: "1 kg", stock: 35,
    image: "https://placehold.co/400x300?text=Jeruk+Pontianak",
    desc: "Jeruk manis khas Pontianak, segar dan banyak air." },
  { id: 9, name: "Pisang Ambon", category: "buah", price: 18000, promo: false,
    unit: "1 sisir", stock: 22,
    image: "https://placehold.co/400x300?text=Pisang+Ambon",
    desc: "Pisang ambon matang alami, manis dan lembut." },
  { id: 10, name: "Mangga Harum Manis", category: "buah", price: 30000, promo: false,
    unit: "1 kg", stock: 18,
    image: "https://placehold.co/400x300?text=Mangga",
    desc: "Mangga harum manis matang pohon." },
  { id: 11, name: "Air Mineral Galon 19 L", category: "minuman", price: 21000, promo: false,
    unit: "galon", stock: 40,
    image: "https://placehold.co/400x300?text=Air+Mineral",
    desc: "Isi ulang galon air minum isi ulang, higienis." },
  { id: 12, name: "Teh Celup 25 pcs", category: "minuman", price: 12000, promo: false,
    unit: "25 kantong", stock: 55,
    image: "https://placehold.co/400x300?text=Teh+Celup",
    desc: "Teh celup harum, pas untuk teman sarapan." }
];

// Format rupiah
const formatRupiah = (n) => "Rp " + n.toLocaleString("id-ID");
```

7.3 `js/products.js` — Render + Search + Filter + Sort

```
document.addEventListener("DOMContentLoaded", () => {
  const listEl     = document.getElementById("productList");
  const emptyEl    = document.getElementById("emptyMessage");
  const searchEl   = document.getElementById("searchInput");
  const catEl      = document.getElementById("categoryFilter");
  const sortEl     = document.getElementById("sortSelect");

  // Ambil ?cat=... dari URL (dipakai link kategori di beranda)
  const params = new URLSearchParams(location.search);
  if (params.get("cat")) catEl.value = params.get("cat");

  let state = { q: "", cat: catEl.value || "all", sort: "default" };

  function cardHTML(p) {
    return `
      <article class="card">
        <img src="${p.image}" alt="${p.name}" class="card__img" loading="lazy">
        <div class="card__body">
          <span class="card__cat">${p.category}</span>
          <h3 class="card__title">${p.name}</h3>
          <p class="card__price">${formatRupiah(p.price)} <small>/ ${p.unit}</small></p>
          ${p.promo ? '<span class="badge-promo">🔥 Promo</span>' : ''}
          <a href="product-detail.html?id=${p.id}" class="btn btn--block">Lihat Detail</a>
        </div>
      </article>`;
  }

  function render() {
    let items = PRODUCTS.filter(p =>
      (state.cat === "all" || p.category === state.cat) &&
      p.name.toLowerCase().includes(state.q)
    );

    if (state.sort === "low")   items.sort((a, b) => a.price - b.price);
    if (state.sort === "high")  items.sort((a, b) => b.price - a.price);
    if (state.sort === "name")  items.sort((a, b) => a.name.localeCompare(b.name));

    listEl.innerHTML = items.map(cardHTML).join("");
    emptyEl.style.display = items.length ? "none" : "block";
  }

  searchEl.addEventListener("input", e => { state.q = e.target.value.toLowerCase(); render(); });
  catEl.addEventListener("change",  e => { state.cat = e.target.value; render(); });
  sortEl.addEventListener("change", e => { state.sort = e.target.value; render(); });

  render();
});
```

> 💡 **Beranda:** Tampilkan 4 produk unggulan dengan menambahkan `js/products.js` versi sederhana yang hanya me-render `PRODUCTS.filter(p => p.promo).slice(0,4)` ke `#featuredProducts`.

BAB 8 — Halaman Detail Produk (product-detail.html + product-detail.js)

8.1 Struktur Halaman

```
<main class="container">
  <a href="products.html">← Kembali ke produk</a>
  <div class="detail" id="productDetail" style="margin-top:1rem;"></div>
</main>
```

CSS:

```
.detail {
  display: grid; grid-template-columns: 1fr; gap: 1.5rem;
  background: var(--white); border: 1px solid var(--border);
  border-radius: var(--radius); padding: 1.5rem; box-shadow: var(--shadow);
}
.detail img { border-radius: var(--radius); }
.detail .price { color: var(--primary); font-size: 1.6rem; font-weight: 700; }
.qty-box { display: flex; align-items: center; gap: .5rem; margin: 1rem 0; }
.qty-box input { width: 70px; padding: .5rem; text-align: center; border: 1px solid var(--border); border-radius: 8px; }

@media (min-width: 768px) {
  .detail { grid-template-columns: 1fr 1fr; align-items: start; }
}
```

8.2 `js/product-detail.js`

```
document.addEventListener("DOMContentLoaded", () => {
  const id = Number(new URLSearchParams(location.search).get("id"));
  const p  = PRODUCTS.find(item => item.id === id);
  const el = document.getElementById("productDetail");

  if (!p) {
    el.innerHTML = "<p>Produk tidak ditemukan.</p>";
    return;
  }

  el.innerHTML = `
    <img src="${p.image}" alt="${p.name}">
    <div>
      <span class="card__cat">${p.category}</span>
      <h1>${p.name}</h1>
      <p class="price">${formatRupiah(p.price)} <small>/ ${p.unit}</small></p>
      <p><strong>Stok:</strong> ${p.stock > 0 ? "Tersedia" : "Habis"}</p>
      <p>${p.desc}</p>
      <div class="qty-box">
        <label for="qty">Jumlah:</label>
        <input type="number" id="qty" value="1" min="1" max="${p.stock}">
      </div>
      <button class="btn" id="addToCart">🛒 Tambah ke Keranjang</button>
    </div>`;

  document.getElementById("addToCart").addEventListener("click", () => {
    const qty = Math.max(1, Number(document.getElementById("qty").value) || 1);
    Cart.add(p.id, qty);          // fungsi dari cart.js
    alert(`"${p.name}" (${qty} ${p.unit.split(" ")[0]}) ditambahkan ke keranjang!`);
  });
});
```

Jangan lupa sertakan script di halaman ini:

```
<script src="js/data.js"></script>
<script src="js/cart.js"></script>
<script src="js/product-detail.js"></script>
<script src="js/app.js"></script>
```

BAB 9 — Keranjang Belanja (cart.html + cart.js)

9.1 Struktur Halaman

```
<main class="container">
  <h1>Keranjang Belanja</h1>
  <div class="table-wrap">
    <table id="cartTable">
      <thead>
        <tr><th>Produk</th><th>Harga</th><th>Jumlah</th><th>Subtotal</th><th></th></tr>
      </thead>
      <tbody id="cartItems"></tbody>
    </table>
  </div>
  <div id="cartEmpty" style="display:none;">😕 Keranjang masih kosong. <a href="products.html" style="color:var(--
primary);">Belanja dulu →</a></div>

  <div class="cart-summary" id="cartSummary">
    <p>Subtotal: <strong id="subtotal">Rp 0</strong></p>
    <p>Ongkir (area Pontianak): <strong id="shipping">Rp 10.000</strong></p>
    <hr>
    <p>Total: <strong id="total" style="color:var(--primary); font-size:1.3rem;">Rp 0</strong></p>
    <a href="checkout.html" class="btn btn--block">Lanjut ke Checkout →</a>
  </div>
</main>
```

CSS tambahan:

```
#cartTable { width: 100%; border-collapse: collapse; background: var(--white); }
#cartTable th, #cartTable td { padding: .75rem; border-bottom: 1px solid var(--border); text-align: left; }
#cartTable th { background: var(--primary); color: #fff; }
.cart-summary {
  max-width: 400px; margin-left: auto; background: var(--white);
  border: 1px solid var(--border); border-radius: var(--radius);
  padding: 1.25rem; margin-top: 1.5rem;
}
.qty-input { width: 60px; padding: .4rem; border: 1px solid var(--border); border-radius: 6px; }
.btn-remove { background: #fee2e2; color: #dc2626; border: none;
  border-radius: 6px; padding: .35rem .7rem; }
```

9.2 `js/cart.js` — Logika Lengkap dengan localStorage

```
const Cart = {
  get() {
    return JSON.parse(localStorage.getItem("cart")) || [];
  },
  save(items) {
    localStorage.setItem("cart", JSON.stringify(items));
    updateBadge();
  },
  add(productId, qty = 1) {
    const items = this.get();
    const found = items.find(i => i.id === productId);
    if (found) found.qty += qty;
    else items.push({ id: productId, qty });
    this.save(items);
  },
  update(productId, qty) {
    const items = this.get();
    const item  = items.find(i => i.id === productId);
    if (item) item.qty = Math.max(1, qty);
    this.save(items);
  },
  remove(productId) {
    this.save(this.get().filter(i => i.id !== productId));
  },
  clear() {
    localStorage.removeItem("cart");
    updateBadge();
  },
  count() {
    return this.get().reduce((sum, i) => sum + i.qty, 0);
  },
  subtotal() {
    return this.get().reduce((sum, i) => {
      const p = PRODUCTS.find(p => p.id === i.id);
      return sum + (p ? p.price * i.qty : 0);
    }, 0);
  }
};

const SHIPPING_COST = 10000; // ongkir flat Pontianak

function updateBadge() {
  const badge = document.getElementById("cartBadge");
  if (badge) badge.textContent = Cart.count();
}
```

9.3 Render Halaman `cart.html`

```
document.addEventListener("DOMContentLoaded", () => {
  const itemsEl = document.getElementById("cartItems");
  if (!itemsEl) return; // bukan halaman cart

  function renderCart() {
    const items = Cart.get();

    document.getElementById("cartEmpty").style.display = items.length ? "none" : "block";
    document.getElementById("cartTable").parentElement.style.display = items.length ? "block" : "none";
    document.getElementById("cartSummary").style.display = items.length ? "block" : "none";

    itemsEl.innerHTML = items.map(i => {
      const p = PRODUCTS.find(p => p.id === i.id);
      if (!p) return "";
      return `
        <tr>
          <td><img src="${p.image}" alt="${p.name}" style="width:50px;height:50px;object-fit:cover;display:inline-
block;vertical-align:middle;margin-right:.5rem;">${p.name}</td>
          <td>${formatRupiah(p.price)}</td>
          <td><input type="number" class="qty-input" min="1" value="${i.qty}" data-id="${p.id}"></td>
          <td>${formatRupiah(p.price * i.qty)}</td>
          <td><button class="btn-remove" data-id="${p.id}">Hapus</button></td>
        </tr>`;
    }).join("");

    const sub = Cart.subtotal();
    document.getElementById("subtotal").textContent = formatRupiah(sub);
    document.getElementById("total").textContent    = formatRupiah(sub + SHIPPING_COST);
  }

  // Event: ubah jumlah / hapus (event delegation)
  document.getElementById("cartItems").addEventListener("input", e => {
    if (e.target.matches(".qty-input")) {
      Cart.update(Number(e.target.dataset.id), Number(e.target.value));
      renderCart();
    }
  });
  document.getElementById("cartItems").addEventListener("click", e => {
    if (e.target.matches(".btn-remove")) {
      Cart.remove(Number(e.target.dataset.id));
      renderCart();
    }
  });

  renderCart();
});
updateBadge();
```

BAB 10 — Checkout (checkout.html + validation.js)

10.1 Struktur Halaman

```
<main class="container">
  <h1>Checkout</h1>
  <div class="checkout-grid">

    <!-- Form data pembeli -->
    <form id="checkoutForm" novalidate>
      <h2>Data Pembeli</h2>
      <div class="form-group">
        <label for="nama">Nama Lengkap *</label>
        <input type="text" id="nama" name="nama">
        <small class="error-msg">Nama minimal 3 karakter.</small>
      </div>
      <div class="form-group">
        <label for="email">Email *</label>
        <input type="email" id="email" name="email">
        <small class="error-msg">Format email tidak valid.</small>
      </div>
      <div class="form-group">
        <label for="hp">No. WhatsApp *</label>
        <input type="tel" id="hp" name="hp" placeholder="08xxxxxxxxx">
        <small class="error-msg">Nomor harus 10–13 digit angka.</small>
      </div>
      <div class="form-group">
        <label for="alamat">Alamat Pengiriman *</label>
        <textarea id="alamat" name="alamat" rows="3" placeholder="Jalan, nomor rumah, kecamatan, kelurahan"></textarea>
        <small class="error-msg">Alamat wajib diisi (min. 10 karakter).</small>
      </div>
      <div class="form-group">
        <label for="bayar">Metode Pembayaran *</label>
        <select id="bayar" name="bayar">
          <option value="">-- Pilih --</option>
          <option value="cod">Bayar di Tempat (COD)</option>
          <option value="transfer">Transfer Bank</option>
          <option value="ewallet">E-Wallet (QRIS)</option>
        </select>
        <small class="error-msg">Pilih metode pembayaran.</small>
      </div>
      <button type="submit" class="btn btn--block">✅ Buat Pesanan</button>
    </form>

    <!-- Ringkasan pesanan -->
    <aside>
      <h2>Ringkasan Pesanan</h2>
      <div id="orderSummary" style="background:#fff; border:1px solid var(--border); border-radius:10px; padding:1rem;">
</div>
    </aside>
  </div>

  <div id="successBox" style="display:none;"></div>
</main>
```

CSS:

```
.checkout-grid { display: grid; grid-template-columns: 1fr; gap: 1.5rem; }
#checkoutForm, aside { background: var(--white); border: 1px solid var(--border);
  border-radius: var(--radius); padding: 1.5rem; }
@media (min-width: 768px) {
  .checkout-grid { grid-template-columns: 3fr 2fr; align-items: start; }
}
```

10.2 `js/validation.js`

```
document.addEventListener("DOMContentLoaded", () => {
  const form = document.getElementById("checkoutForm");
  if (!form) return;

  // --- Ringkasan pesanan ---
  function renderSummary() {
    const box = document.getElementById("orderSummary");
    const items = Cart.get();
    if (!items.length) {
      box.innerHTML = '<p>Keranjang kosong. <a href="products.html" style="color:var(--primary)">Belanja dulu →</a></p>';
      return;
    }
    const rows = items.map(i => {
      const p = PRODUCTS.find(p => p.id === i.id);
      return `<p>${p.name} × ${i.qty} — <strong>${formatRupiah(p.price * i.qty)}</strong></p>`;
    }).join("");
    const sub = Cart.subtotal();
    box.innerHTML = rows + `<hr>
      <p>Subtotal: <strong>${formatRupiah(sub)}</strong></p>
      <p>Ongkir: <strong>${formatRupiah(SHIPPING_COST)}</strong></p>
      <p>Total: <strong style="color:var(--primary);font-size:1.2rem;">${formatRupiah(sub + SHIPPING_COST)}</strong>
</p>`;
  }
  renderSummary();

  // --- Validasi ---
  const rules = {
    nama:    v => v.trim().length >= 3,
    email:   v => /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(v),
    hp:      v => /^0[0-9]{9,12}$/.test(v.trim()),
    alamat:  v => v.trim().length >= 10,
    bayar:   v => v !== ""
  };

  function validateField(input) {
    const ok = rules[input.name](input.value);
    input.closest(".form-group").classList.toggle("invalid", !ok);
    return ok;
  }

  // Validasi realtime saat keluar dari input
  form.querySelectorAll("input, textarea, select").forEach(el => {
    el.addEventListener("blur", () => validateField(el));
  });

  form.addEventListener("submit", e => {
    e.preventDefault();

    // Cek semua field
    let valid = true;
    form.querySelectorAll("input, textarea, select").forEach(el => {
      if (!validateField(el)) valid = false;
    });

    // Cek keranjang tidak kosong
    if (!Cart.get().length) {
      alert("Keranjang masih kosong!");
      valid = false;
    }

    if (!valid) return;

    // --- Simulasi pesanan berhasil (tanpa backend) ---
    const total = Cart.subtotal() + SHIPPING_COST;
    const orderNo = "GS-" + Date.now().toString().slice(-8);

    form.style.display = "none";
    const box = document.getElementById("successBox");
    box.style.display = "block";
    box.innerHTML = `
```

```
      <div class="alert-success">
        <h2>🎉 Pesanan Berhasil!</h2>
        <p>Nomor pesanan: <strong>#${orderNo}</strong></p>
        <p>Total belanja: <strong>${formatRupiah(total)}</strong></p>
        <p>Tim kami akan menghubungi Anda via WhatsApp untuk konfirmasi pengiriman.</p>
        <a href="products.html" class="btn">Belanja Lagi</a>
      </div>`;

    Cart.clear();   // kosongkan keranjang
    box.scrollIntoView({ behavior: "smooth" });
  });
});
```

⚠️ **Penting:** Ini simulasi. Website statis (tanpa backend) tidak boleh menyimpan data kartu/kredensial. Untuk pembayaran asli, butuh payment gateway (Midtrans/Xendit) + backend — di luar cakupan modul ini.

BAB 11 — Halaman Kontak & Tentang Kami

11.1 `contact.html`

```
<main class="container">
  <h1>Hubungi Kami</h1>
  <div class="contact-grid">
    <!-- Info toko -->
    <div>
      <h2>Info Toko</h2>
      <p>📍 Jl. Contoh Raya No. 12, Pontianak Kota, Kalimantan Barat</p>
      <p>📱 WhatsApp: 0812-xxxx-xxxx</p>
      <p>✉️ email@grosirsegarpontianak.id</p>
      <p>🕐 Setiap hari: 07.00 – 21.00 WIB</p>
      <iframe src="https://maps.google.com/maps?q=Pontianak&t=&z=13&ie=UTF8&iwloc=&output=embed"
        width="100%" height="250" style="border:0; border-radius:10px; margin-top:1rem;" loading="lazy"></iframe>
    </div>

    <!-- Form pesan -->
    <form id="contactForm" novalidate style="background:#fff; border:1px solid var(--border); border-radius:10px;
padding:1.5rem;">
      <h2>Kirim Pesan</h2>
      <div class="form-group">
        <label for="cNama">Nama *</label>
        <input type="text" id="cNama" required>
        <small class="error-msg">Nama wajib diisi.</small>
      </div>
      <div class="form-group">
        <label for="cEmail">Email *</label>
        <input type="email" id="cEmail" required>
        <small class="error-msg">Format email tidak valid.</small>
      </div>
      <div class="form-group">
        <label for="cPesan">Pesan *</label>
        <textarea id="cPesan" rows="5" required></textarea>
        <small class="error-msg">Pesan minimal 10 karakter.</small>
      </div>
      <button type="submit" class="btn btn--block">Kirim Pesan</button>
      <div id="contactSuccess" class="alert-success" style="display:none; margin-top:1rem;">
        ✅ Pesan terkirim! Terima kasih atas pesan Anda.
      </div>
    </form>
  </div>
</main>
```

CSS + JS singkat:

```
.contact-grid { display: grid; grid-template-columns: 1fr; gap: 1.5rem; }
@media (min-width: 768px) { .contact-grid { grid-template-columns: 1fr 1fr; } }

document.getElementById("contactForm").addEventListener("submit", e => {
  e.preventDefault();
  const nama = document.getElementById("cNama");
  const email = document.getElementById("cEmail");
  const pesan = document.getElementById("cPesan");
  let ok = true;
  [[nama, nama.value.trim().length >= 3],
   [email, /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email.value)],
   [pesan, pesan.value.trim().length >= 10]].forEach(([el, valid]) => {
    el.closest(".form-group").classList.toggle("invalid", !valid);
    if (!valid) ok = false;
  });
  if (ok) {
    document.getElementById("contactSuccess").style.display = "block";
    e.target.reset();
  }
});
```

11.2 `about.html`

```
<main class="container">
  <h1>Tentang Kami</h1>
  <p><strong>Grosir Segar Pontianak</strong> adalah toko kelontong online yang melayani kebutuhan sembako,
  sayur, buah, dan kebutuhan harian warga Pontianak sejak 2020. Kami mengutamakan kesegaran produk,
  harga terjangkau, dan pengiriman tepat waktu.</p>

  <div class="feature-grid" style="margin-top:1.5rem;">
    <div class="card"><div class="card__body"><h3>Visi</h3><p>Menjadi toko grocery online terpercaya di Kalimantan Barat.
</p></div></div>
    <div class="card"><div class="card__body"><h3>Misi</h3><p>Menyediakan produk segar dengan harga terjangkau dan
pelayanan ramah.</p></div></div>
    <div class="card"><div class="card__body"><h3>Nilai Kami</h3><p>Jujur, segar, tepat waktu, dan dekat dengan
pelanggan.</p></div></div>
  </div>

  <h2 style="margin-top:2rem;">Layanan Kami</h2>
  <ul style="list-style:disc; padding-left:1.5rem;">
    <li>Pengiriman same-day area Pontianak</li>
    <li>Pesanan grosir untuk warung & katering</li>
    <li>Jaminan produk segar atau uang kembali</li>
    <li>Pembayaran COD, transfer, dan QRIS</li>
  </ul>
</main>
```

BAB 12 — Testing & Debugging (Checklist)

12.1 Uji Responsif

- Buka Chrome → F12 → klik ikon device toolbar (Ctrl+Shift+M).
- Uji ukuran:
  - Mobile: 360×640 (Samsung), 375×667 (iPhone)
  - Tablet: 768×1024 (iPad)
  - Desktop: 1366×768, 1920×1080
- Pastikan: tidak ada scroll horizontal, nav jadi hamburger di HP, grid menyesuaikan, tombol mudah ditekan.

12.2 Uji Fungsional (Alur Pengguna)

- Nav link menuju halaman benar di semua halaman.
- Menu mobile terbuka/tutup saat tombol ☰ diklik.
- Badge keranjang berubah saat tambah produk.
- Search, filter kategori, dan sorting berfungsi.
- Link "Lihat Detail" membawa ke produk yang benar (?id=).
- Ubah jumlah & hapus item di keranjang → total ikut berubah.
- Refresh halaman → keranjang tetap ada (localStorage).
- Checkout dengan form kosong/salah → muncul pesan error.
- Checkout valid → muncul pesanan sukses, keranjang dikosongkan.
- Form kontak validasi berfungsi.

12.3 Uji Teknis

- Buka Console (F12 → Console): tidak ada error merah.
- Tab Network: tidak ada resource 404 (gambar/CSS/JS gagal).
- Jalankan Lighthouse (Chrome DevTools → tab Lighthouse): target Performance ≥ 80, Accessibility ≥ 90.
- Cek semua browser: Chrome, Firefox, Edge.

12.4 Masalah Umum & Solusi

Gejala	Penyebab	Solusi
CSS tidak muncul	Path salah (`/css/...` vs `css/...`)	Pakai path relatif `css/style.css`
JS error "PRODUCTS is not defined"	Urutan script salah	`data.js` harus dimuat pertama
Gambar tidak tampil	Nama file beda (case-sensitive)	Cocokkan persis nama file
Menu mobile tidak muncul	Lupa `classList.toggle("show")`	Cek `#navToggle` & `#navMenu` di `app.js`
Keranjang kosong terus	Belum simpan ke localStorage	Cek `Cart.save()` dipanggil
Konten meluber di HP	Lupa meta viewport	Tambahkan meta viewport

Tambahan `js/app.js` untuk nav mobile + badge:

```
document.addEventListener("DOMContentLoaded", () => {
  // Toggle menu mobile
  const toggle = document.getElementById("navToggle");
  const menu   = document.getElementById("navMenu");
  if (toggle && menu) {
    toggle.addEventListener("click", () => menu.classList.toggle("show"));
  }
  // Badge keranjang
  if (typeof Cart !== "undefined") updateBadge();
});
```

BAB 13 — SEO Dasar & Publish Online

13.1 SEO Dasar (di Setiap Halaman)

- `<title>` unik dan deskriptif, mis. "Katalog Produk - Grosir Segar Pontianak".
- `<meta name="description">` 120–160 karakter per halaman.

- Satu `<h1>` per halaman, heading berurutan (h1 → h2 → h3).
- Semua `<img>` punya atribut `alt`.
- Tambahkan favicon: `<link rel="icon" href="assets/images/logo.png">`.

13.2 Persiapan Sebelum Publish

- Cek path — semua path relatif (`css/`, `js/`, `assets/`), bukan absolut (`/css/`).
- Kompres gambar — gunakan format `.webp` atau JPEG < 100 KB per gambar (tools: tinypng.com).
- Bersihkan console error dan link rusak.
- Buat `README.md` berisi deskripsi proyek, cara menjalankan, dan daftar halaman.

13.3 Cara Menjalankan Lokal

- Cara termudah: klik kanan file `index.html` → Open with Chrome.
- Cara lebih baik (direkomendasikan): pakai extension Live Server di VS Code → klik kanan `index.html` → Open with Live Server (reload otomatis saat file berubah).

13.4 Publish GRATIS — Pilih Salah Satu

Opsi A — Netlify (paling mudah, tanpa Git)

1. Buka https://app.netlify.com/drop.
2. Drag & drop seluruh folder `grocery-pontianak` ke halaman tersebut.
3. Tunggu beberapa detik → dapat URL publik seperti `https://random-name.netlify.app`.
4. Selesai! Bisa diganti nama site di Site settings → Change site name.

Opsi B — GitHub Pages (butuh akun GitHub)

1. Buat akun GitHub → buat repository baru (mis. `grocery-pontianak`), centang Add a README.
2. Upload semua file proyek ke repository (drag & drop di web, atau via Git):

```
git init
git add .
git commit -m "Initial commit: grocery store pontianak"
git branch -M main
git remote add origin https://github.com/USERNAME/grocery-pontianak.git
git push -u origin main
```

3. Buka tab Settings → Pages.
4. Di Source: pilih branch `main`, folder `/ (root)` → Save.
5. Tunggu 1–2 menit → website live di `https://USERNAME.github.io/grocery-pontianak/`.

Opsi C — Vercel

1. Push kode ke GitHub (seperti Opsi B langkah 2).
2. Buka https://vercel.com/new → Import repository → klik Deploy.
3. Dapat URL publik gratis.

13.5 Verifikasi Setelah Publish

- Buka URL publik di HP asli (bukan hanya DevTools).
- Uji ulang: nav, search, tambah ke keranjang, checkout.
- Pastikan semua gambar/CSS/JS termuat (cek Network tab).
- Bagikan URL ke rekan untuk review.

📌 Ringkasan Urutan Pengerjaan

1. Buat folder + file sesuai struktur (Bab 2).
2. Buat template HTML dasar → copy ke semua halaman (Bab 3).
3. Tulis CSS global: variabel, header, footer, komponen (Bab 4).
4. Tulis `responsive.css` (Bab 5).
5. Susun Beranda (Bab 6).
6. Isi `data.js` + katalog produk dengan search/filter (Bab 7).
7. Detail produk via URL `?id=` (Bab 8).
8. Keranjang + `localStorage` (Bab 9).
9. Checkout + validasi form (Bab 10).
10. Kontak + Tentang Kami (Bab 11).
11. Testing di 3 ukuran layar + 3 browser (Bab 12).
12. SEO checklist → Upload ke Netlify/GitHub Pages (Bab 13).

> ✅ **Batasan yang harus diketahui:** Versi ini website *frontend only* — pembayaran, login, dan database membutuhkan backend yang dipelajari di modul lanjutan.

Modul Materi: Desain Website Grocery Store Pontianak
HTML5 + CSS3 + JavaScript Vanilla — Responsive — Dari Nol Sampai Publish
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